1. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of the problem

Highest Relevance

The question requires judging whether Kevin's statement is correct and explaining why. This directly corresponds to the ability to identify whether a mathematical conclusion is reasonable within a real-world context. Kevin erroneously assumes that similar percentages imply a larger absolute number, ignoring the significant difference in the total number of households between France and Norway. The task is to point out why his conclusion does not hold in context.

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2. Construct and communicate explanations and arguments in the context of the problem

Highest Relevance

The question explicitly asks to "explain why this statement is incorrect. Give a reason for your answer." This requires students to construct a clear argument, using data to illustrate Kevin's error (e.g., by calculating absolute subscription numbers or comparing total households), thereby exercising the ability to communicate and explain within the problem's context.

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3. Interpret a mathematical result back into the real world context

High Relevance

Students need to reinterpret the percentage data from the table (e.g., 15.4% and 42.7%) as actual numbers of households, understanding that "more" refers to absolute quantity, not relative proportion. This demonstrates the ability to translate a mathematical result back into its real-world setting.

4. Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations

High Relevance

The question involves two types of data representation: percentages (TV ownership percentage, subscription percentage) and absolute numbers (households owning TVs). Students must compare these representations and evaluate Kevin's mistake of relying solely on percentages while ignoring absolute figures, thereby understanding the relationship between different data formats.

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5. Understand the relationship between the context of the problem and representation of the mathematical solution

Medium-High Relevance

Understanding the link between the problem's context (different country sizes) and the mathematical representation (percentages) is key. Kevin's error stems from disconnecting the context from the representation. A correct understanding requires recognizing that percentages depend on a base value, and when base values differ, percentages cannot be directly compared.

6. Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions

Medium Relevance

This question highlights a limitation of the mathematical concept of percentages: they only reflect proportional relationships, not absolute magnitudes. Students need to recognize this limitation to correctly explain Kevin's error.

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7. Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied

Medium Relevance

This requires applying real-world knowledge (i.e., the difference in total households between countries) to inform the interpretation of percentages. While the question doesn't require adjusting a calculated result, making a correct judgment depends on considering this real-world context.

1. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of the problem

相关性最高

题目要求判断Kevin的说法是否正确，并解释原因。这直接对应识别数学结论在现实背景下是否合理的能力。Kevin错误地将百分比相近等同于绝对数量更多，而忽略了法国和挪威总家庭数的巨大差异，因此需要指出其结论在现实背景下不成立。

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2. Construct and communicate explanations and arguments in the context of the problem

相关性最高

题目明确要求“解释为什么这个说法是错误的，并给出理由”。这需要学生构建一个清晰的论证，结合数据说明Kevin的错误所在（例如，通过计算绝对订阅量或比较总家庭数），从而锻炼在问题背景下沟通和解释的能力。

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3. Interpret a mathematical result back into the real world context

相关性高

学生需要将表格中的百分比数据（如15.4%和42.7%）重新解释为实际家庭数量，理解“更多”指的是绝对数值而非相对比例。这体现了将数学结果放回现实背景进行解读的能力。

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4. Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations

相关性高

题目涉及两种数据表示形式：百分比（拥有电视的家庭占比、订阅占比）和绝对数量（拥有电视的家庭数）。学生需要比较这些表示，并评估Kevin仅依赖百分比而忽略绝对数量的错误，从而理解不同表示之间的关系。

5. Understand the relationship between the context of the problem and representation of the mathematical solution

相关性中等偏高

理解问题背景（国家规模不同）与数学表示（百分比）之间的联系是关键。Kevin的错误在于割裂了背景与表示，而正确理解需要认识到百分比依赖于基数，基数不同则百分比不能直接比较。

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6. Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions

相关性中等

本题揭示了百分比这一数学概念的局限性：百分比只能反映比例关系，不能直接代表绝对大小。学生需要认识到这一局限，才能正确解释错误。

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7. Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied

相关性中等

这里需要运用现实世界知识（即不同国家的总家庭数差异）来调整对百分比的理解。虽然题目没有要求调整计算结果，但正确判断依赖于对现实背景的考量。